



Measurement WorkFlow Remote

User manual for browser-based scan setup, equipment detection, run control, and transmission-spectrum review.

Purpose

Configure and start Measurement WorkFlow scans from a phone or desktop browser while the lab PC remains the experiment control point.

Supported Workflows

Oscilloscope acquisition and Laser Doppler Vibrometer measurement workflows.

Primary Users

Experimental vibration, dynamics, and acoustics researchers.

Desktop Authority

The desktop application remains responsible for hardware communication and experiment execution.

Document type

User Manual

Revision

May 26, 2026

Software

Measurement WorkFlow Remote

Prepared for

APTM Lab measurement workflow

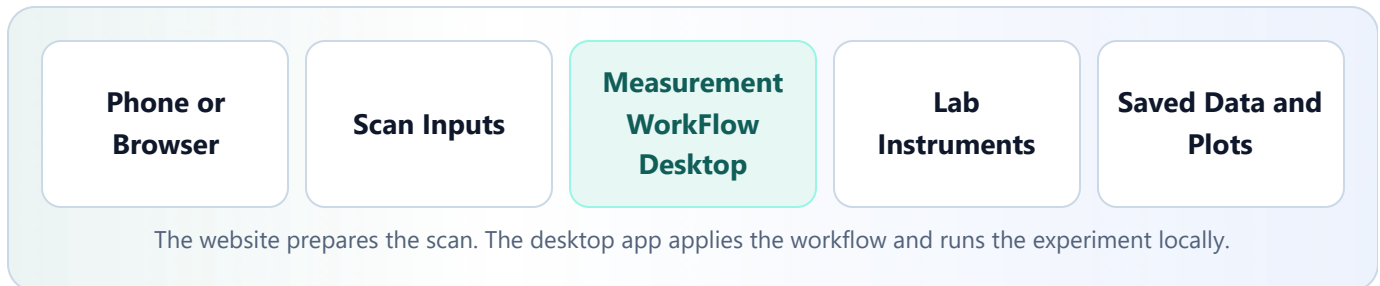
| Contents

1. System Overview	01	2. Requirements	02
3. Login and Connection	03	4. Choose Scan Workflow	04
5. Setup and Detect	05	6. Scan Inputs	06
7. Run Control	07	8. Transmission Spectrum View	08
9. Troubleshooting	09	10. Field Reference	10

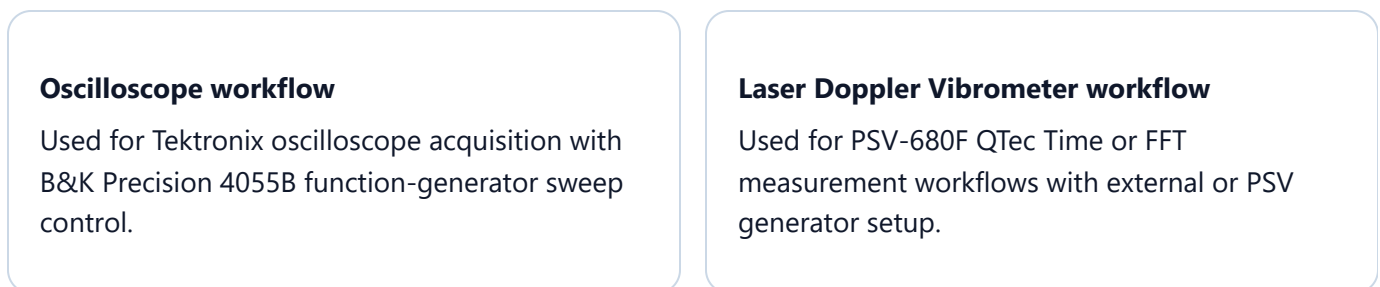
Manual scope: This document covers the browser website. The desktop application remains the authority for hardware communication, readiness checks, plotting, logging, and equipment shutdown.

| Browser Interface For Scan Setup

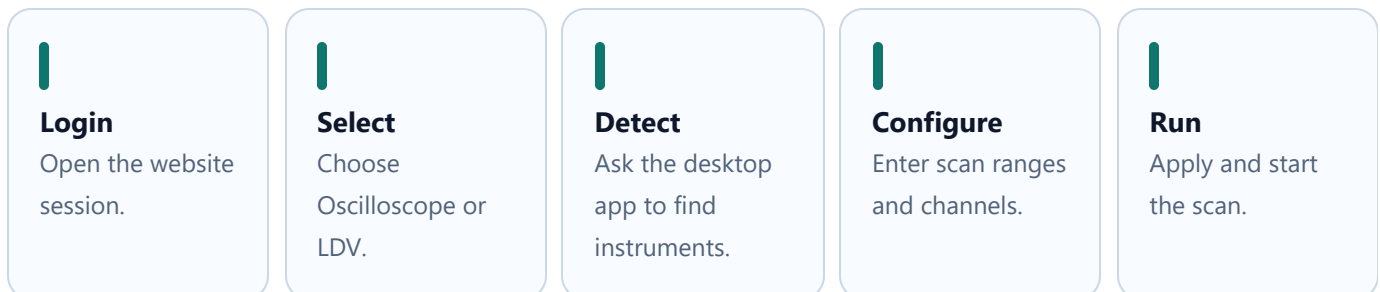
Measurement WorkFlow Remote is the browser interface for preparing scan inputs for the Measurement WorkFlow desktop application. It is designed for phone and desktop browsers. The desktop app remains responsible for hardware communication, readiness checks, experiment execution, data saving, and plotting.



Supported acquisition families



Typical workflow



| What Must Be Ready

Lab PC

- Measurement WorkFlow desktop application installed and able to run scans locally.
- Required instruments connected and visible to the desktop app.
- Output folders available for saving experiment data.
- Internet access from the lab PC when the browser website is used outside the local machine.

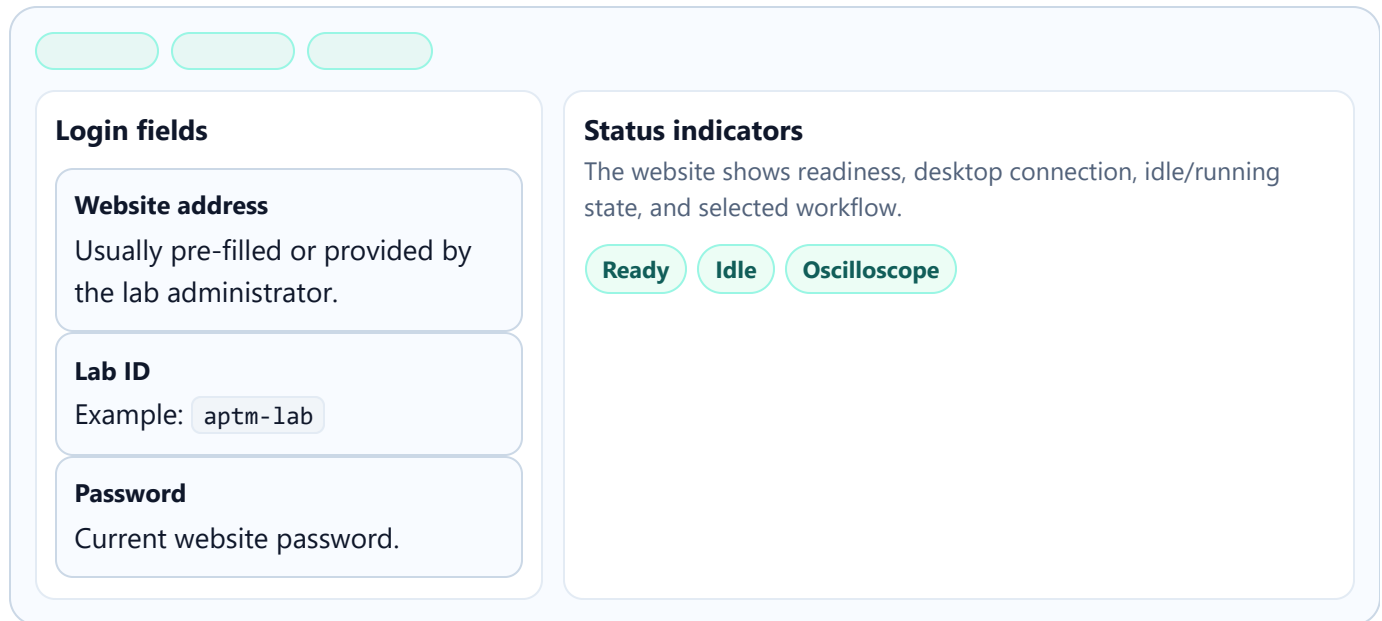
Website

- A browser on a phone, tablet, or computer.
- Website link provided by the lab administrator.
- Lab ID.
- Current website password.

Before running: confirm that the physical setup is safe, cables are connected correctly, the excitation path is ready, and no one is adjusting the mechanical system.

Sign In To The Website

The login page identifies the lab session and allows the browser to send scan setup commands to the desktop application. The browser may remember the session locally so the page can resume later.



The login page interface is divided into two main sections: 'Login fields' and 'Status indicators'. The 'Login fields' section contains three input fields: 'Website address' (with a description: 'Usually pre-filled or provided by the lab administrator.'), 'Lab ID' (with an example: 'aptm-lab'), and 'Password' (with a description: 'Current website password.'). The 'Status indicators' section shows the website's readiness, desktop connection, idle/running state, and selected workflow. It includes three buttons: 'Ready', 'Idle', and 'Oscilloscope'.

Login procedure

1. Open the Measurement WorkFlow Remote website.
2. Confirm the website address if the field is visible.
3. Enter the Lab ID.
4. Enter the website password.
5. Select **Login**.

Clear Saved Login

Use **Clear Saved Login** when the password changes, the website address changes, or another lab member needs to log in with their current credentials.

Login duration: If the website reports an expired login, clear the saved login and sign in again.

Open The Correct Desktop Scan Stage

After login, the Home page shows two acquisition families. Selecting one sends an **Apply** command to the desktop app so the desktop scan page opens the matching workflow.

Oscilloscope

Use this workflow for Tektronix oscilloscope acquisition with B&K Precision 4055B sweep control.

Laser Doppler Vibrometer

Use this workflow for PSV-680F QTec Time or FFT acquisition and LDV transmission-spectrum analysis.

Stepper navigation

STAGE	USE
Home	Choose Oscilloscope or Laser Doppler Vibrometer.
Setup and Detect	Enter output folders and detect available instruments.
Scan Inputs	Enter sweep/channel settings and run the scan.

Prepare Instruments Before Entering Scan Inputs

LDV setup fields

FIELD	DESCRIPTION
Output folder	Folder where scan data should be saved. If left blank on the website, the default data folder is used.
Mass count	Number of masses or response points expected in the LDV dataset.
Experiment settle (s)	Wait time after a frequency or excitation-state change before acquisition begins.
Generator Mode	External Function Generator or PSV Internal Generator. The PSV internal generator mode does not use phase input.
Scan Type	Frequency/Amplitude/Phase or ϵ and δ_0 sweep.
External Generator VISA Resources	Detected or manually pasted B&K Precision 4055B VISA resource strings.

Oscilloscope setup fields

FIELD	DESCRIPTION
Output folder	Folder where oscilloscope scan data should be saved.
Oscilloscope VISA Resources	Detected or manually pasted Tektronix oscilloscope VISA resources.
Function Generator VISA Resources	Detected or manually pasted B&K Precision 4055B function-generator resources.
Generator Mode	External Function Generator. Oscilloscope workflow uses external generator control.
Scan Type	Frequency/Amplitude/Phase or ϵ and δ_0 sweep.

Detect buttons

Detect Oscilloscopes

Asks the desktop app to detect Tektronix resources.

Detect Function Generators

Asks the desktop app to detect B&K Precision 4055B resources.

Detect Both

Runs oscilloscope and function-generator detection from one button.

Detection is not continuous. The website sends a detect command when you press a detect button. The desktop app performs detection and later reports the result back to the website.

Enter The Scan Definition

Standard function-generator channel input

Each detected B&K Precision 4055B function generator has two channels. The website pages through function generators so only one generator is edited at a time.

CHANNEL MODE	ACTIVE FIELDS	USE CASE
Sweep Frequency	Frequency low, frequency high, frequency step, fixed amplitude, fixed phase.	Frequency response measurements at a constant excitation amplitude.
Sweep Amplitude	Fixed frequency, amplitude low, amplitude high, amplitude step, fixed phase.	Amplitude-dependent response checks.
Sweep Phase	Fixed frequency, fixed amplitude, phase low, phase high, phase step.	Phase excitation studies using external function generators.
Fixed Condition	Fixed frequency, fixed amplitude, fixed phase.	Holding one channel constant while another channel sweeps.

Function-generator output settings

Waveform

Choose the waveform for each function generator. Default is Sine.

CH Coupling and Wave Combine

Configure output behavior when the generator should couple channels or combine waveforms.

ϵ and δ_0 sweep input

This mode keeps the ϵ and δ_0 logic from the desktop workflow while also allowing frequency control. If only one function generator is active, it is selected automatically. If multiple generators are active, choose the target generator.

CONTROL	DESCRIPTION
Amplitude cap Vpp	Maximum voltage cap used by the ϵ/δ_0 logic.
Sweep	Select Sweep Frequency, Sweep ϵ , or Sweep δ_0 .
Same Frequency For CH1 And CH2	When enabled, CH2 follows CH1 frequency inputs.
Divide by step / points	Choose whether the step field represents a direct step size or a number of points.

I Apply, Run, Pause, Stop

The Run Control panel sends direct workflow commands to the desktop app. The desktop app still checks readiness before the experiment starts.

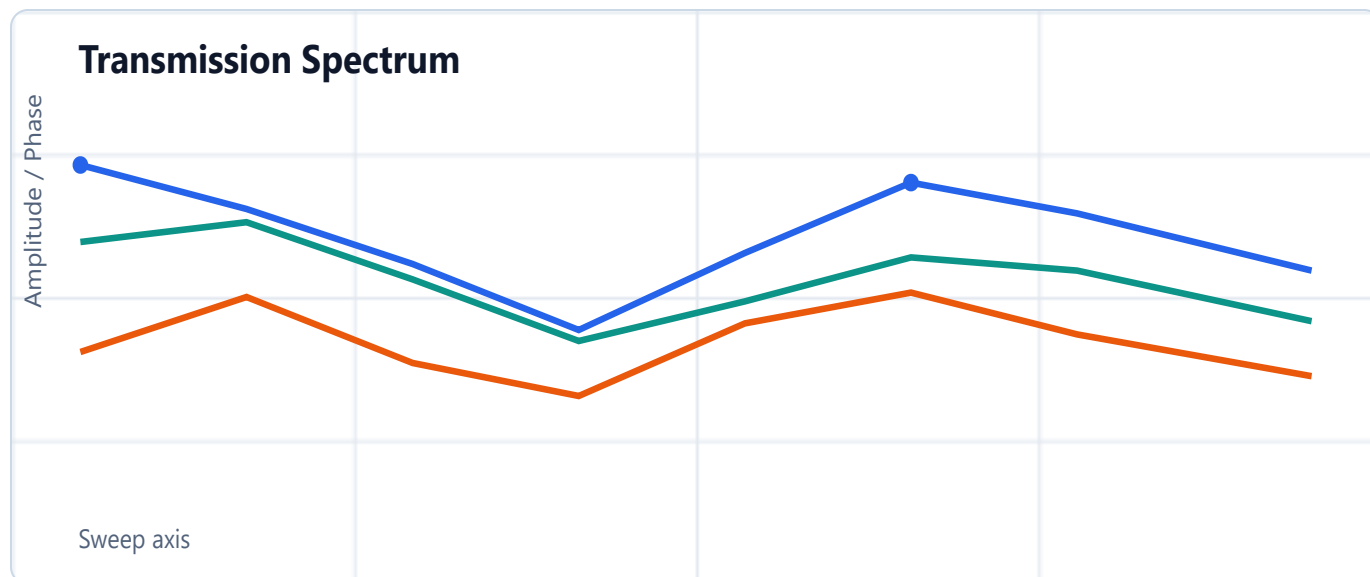
BUTTON	ACTION
Back to Setup	Return to the Setup and Detect stage without running the experiment.
Refresh	Requests the latest status and detected resources from the desktop app.
Apply	Sends current scan inputs to the desktop app without starting the scan.
Apply & Run	Sends current scan inputs and requests scan start.
Pause	Requests a pause/resume style control action from the desktop app.
Stop	Requests the active scan to stop.
Outputs Off	Requests function-generator outputs to turn off.
Reset Inputs	Requests a reset of desktop scan inputs.

Long scans

During a run, the website checks for updated status periodically. If the phone sleeps or the browser is suspended, the desktop experiment can continue. Reopen the website and log in again if needed to view updated status.

Review The Final Plot

After a scan completes and the desktop app reports transmission-spectrum data, the remote website can show the final transmission-spectrum figure. This view is intentionally minimal: it displays the plot only, so it can be reviewed quickly on a phone.



What is displayed

- Amplitude response across completed scan points.
- Phase response between masses when available.
- Mass-driver phase when available.
- Legend entries for reported masses or response points.

Plot source: The plot is generated from the desktop app status payload. If the desktop app has not reported a completed transmission spectrum, the website will continue to show scan controls and status instead of the final plot.

Common Problems

MESSAGE OR BEHAVIOR	LIKELY CAUSE	ACTION
Invalid login	The Lab ID or website password is not correct.	Confirm the current password with the lab administrator, then clear saved login and sign in again.
Failed to fetch	Wrong website address, network issue, or the website service is temporarily unavailable.	Confirm the website address, check internet connection, and try refreshing the page.
Desktop app is not connected	The desktop app is closed or not currently accepting website commands.	Open Measurement WorkFlow on the lab PC and confirm the website connection is enabled.
Detect command sent, but resources do not update	Instrument is not visible to VISA, app is busy, or status has not refreshed yet.	Wait for detection refresh, press Refresh, and verify resources locally on the lab PC.
Website session appears lost during a long experiment	Phone browser slept or the tab was suspended.	Reopen the website, log in if needed, and refresh status. The desktop scan can continue independently.
Transmission spectrum does not appear	The desktop app has not completed or has not reported spectrum data.	Check the desktop app scan status and confirm completed metrics exist.

Quick Reference

Default folders and labels

ITEM	DEFAULT OR LABEL
Default data root	D:\Polytec_Software_Code\Data
Function generator	B&K Precision 4055B
Oscilloscopes	Tektronix MDO 3 Series and Tektronix MDO3024
LDV	PSV-680F QTec Laser Doppler Vibrometer

Command behavior

WEBSITE COMMAND	INCLUDES SCAN PRESET?	DESKTOP BEHAVIOR
Apply	Yes	Applies current website inputs to the desktop app.
Apply & Run	Yes	Applies inputs and requests scan start.
Detect commands	Yes	Uses current workflow and asks desktop app to detect instruments.
Pause, Stop, Outputs Off, Reset	No	Sends direct control command to desktop app.

Measurement WorkFlow Remote is intended to support laboratory measurement setup and monitoring. The desktop Measurement WorkFlow application remains responsible for hardware communication, scan execution, data saving, and final analysis.